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BRAC UNIVERSITY
Department of Computer Science & Engineering
FINAL EXAMINATION, SUMMER 2024
CSE 350: Digital Electronics and Pulse Techniques

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Total Marks: 60 (20×3)
Time: **90 Minutes**

- Total no of questions are **four (4)**. Answer any **three (3)**.
- Figure in bracket [ ] next to each question indicates marks for that question.
- Write your set number on top of your answer sheet
- There is no need to draw circuits in your answer sheet.

Name:	ID:	Section:

Question 1 [20]

For this **Schmitt Trigger** circuit in Figure 1, given: V_{H} = 10 V, V_{L} = -10 V, v_{ref} = 2 V.

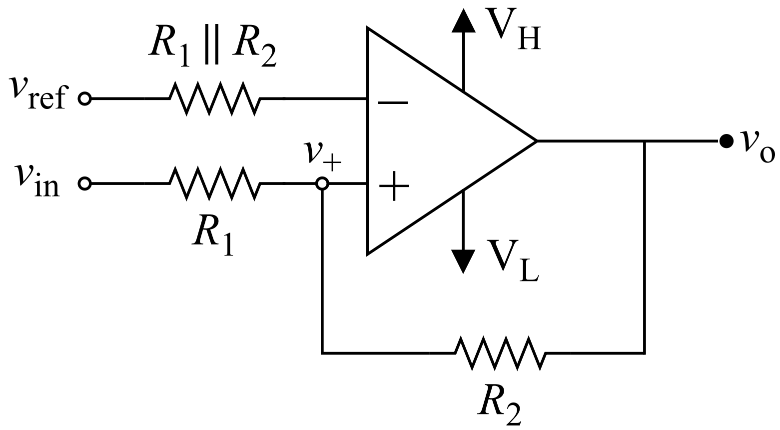

Figure 1: Schmitt Trigger

CO1	(a)	What **type** of Schmitt trigger circuit is this (inverting or non-inverting)?	[2]

For question (b), given: $R_1 = 1\text{k}\Omega$ and $R_2 = 2\text{k}\Omega$.

CO1	(b)	**Draw** the **VTC** (**v_i vs v_o**) with **appropriate labels** (Values of V_{H}, V_{L}, V_{TH}, V_{TL}, Hysteresis Width)	[8]

Consider the same Figure 1. Now for question (c), $V_{\mathrm{TH}} = 5V$, $V_{\mathrm{TL}} = 0V$ and $R_1||R_2 = 4k\Omega$.

CO1	(c)	**Calculate** the value of R_1 and R_2.	[10]

Question 2 [20]

Figure 2 consists of a square wave generator, and Figure 3 depicts a logical function.

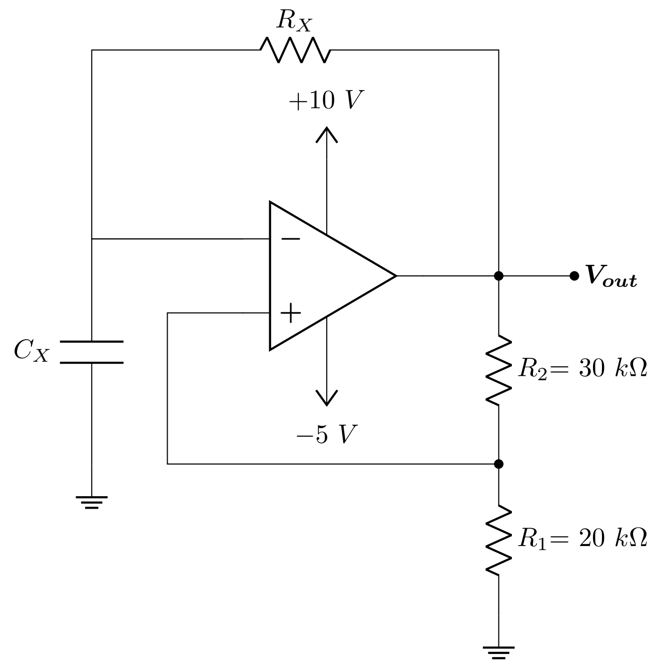

Figure 2: Square Wave Generator

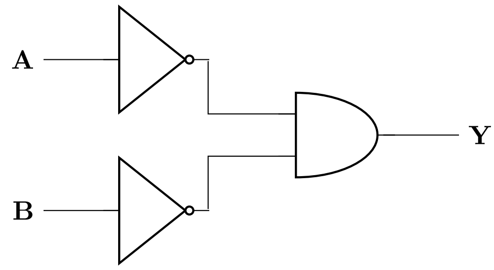

Figure 3: Logic Circuit

Consider the square wave generator in Figure 2.

CO1	(a)	**Determine** the values of V_{TH} *(upper threshold)* and V_{TL}*(lower threshold).*	[2+2]
CO1	(b)	**Determine** the period, time constant, and the duty cycle of the square wave generator, given that the frequency is $100Hz$.	[8]

Consider the logic circuit in Figure 3.

CO1	(a)	**Design** the equivalent CMOS logic circuit for the logic circuit using a combination of NMOS and PMOS transistors. Hence, **write** down the truth table showing the "ON" or "OFF" state of the MOSFETS for each input case.	[8]

Question 3 [20]

The figure below shows the input-output characteristics of a converter. The output of the converter is denoted as V_0 and the input as V_{in}.

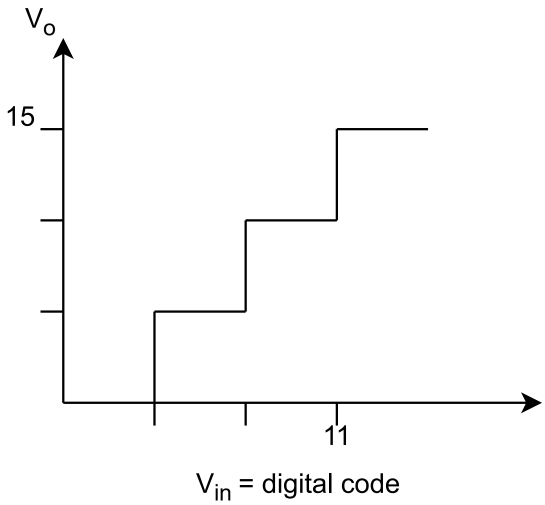

Figure 4

CO3	(a)	**Identify** the converter type and the number of bits used by the converter. Further, **calculate** V_{ref} when $R_F = 2R$.	[2+4]
CO3	(b)	**Draw** the converter with appropriate values from part (a).	[4]
CO3	(c)	**Calculate** all the outputs, V_0, for the converter. Copy the transfer curve in Figure 4 and **write down** the missing input and output values.	[8+2]

Question 4 [20]

The figure shows a 2-bit Flash ADC with $V_{REF} = 10\text{V}$ and its input ranged from 0 to 10V. **Given:** $R_1 = 5\text{k}\Omega$, $R_2 = 10\text{k}\Omega$, $R_3 = 10\text{k}\Omega$, $R_4 = 15\text{k}\Omega$. The two outputs of the ADC are connected to two LEDs.

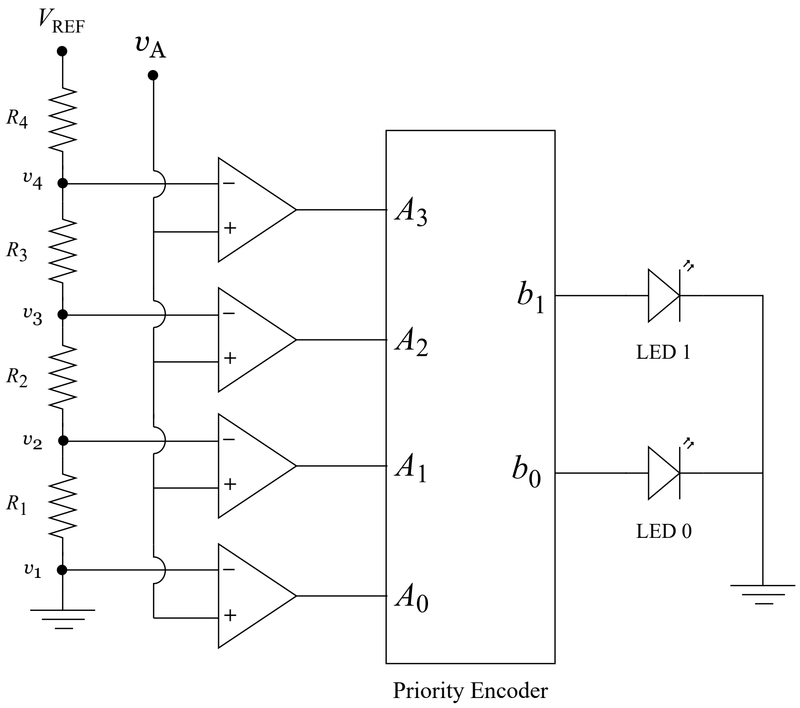

Figure 5: 2-bit flash ADC

Sampled Value (V)	Time (s)
5.00	0.0
7.40	0.5
5.71	3.0
1.22	4.0

Table 1: Sampled values at different time instances

CO3	(a)	**Determine** the quantization range, corresponding digital output, and the states of the LED for the 2-bit Flash ADC shown in Figure 5.	[5+5]
CO3	(b)	Four sample values at four different time instances obtained from an analog signal are shown in Table 1. **Identify** the quantization range for each sampled value. **Determine** the corresponding encoded output, and **identify** the states of the LEDs.	[5+5]